5 *Output token IDs*—The LLM's response in numerical format.
6 *Detokenization*—The conversion of numerical IDs back into text tokens.
7 *Human-readable output*—The final stage where tokens are combined into coherent text for the user.

While tokenization is important, we also need to carefully consider the limitations imposed by various LLM providers regarding the size of the prompts and how we control sections within prompts.

In the following section, we take a closer look at two equally important fundamentals in prompt engineering and LLM interaction: max tokens and control tokens.

MAX TOKENS

In your interactions with popular LLMs, you may have encountered the concept of max tokens, which refers to the maximum number of tokens an LLM can process in a single interaction, including both the input (prompt) and the output (generated text). The user input is tokenized into smaller units, such as words or subwords, and each token counts toward the total limit. For example, if an LLM has a maximum context window of 8,192 tokens and your input consists of 200 tokens, only 7,992 tokens remain for the output. Therefore, it's crucial to consider the length of your input to ensure there are enough tokens left for meaningful responses from the model. Balancing input length and output expectations helps optimize interactions with LLMs and enhances overall performance.

For example, GPT-3 models typically have a limit of 4,096 tokens, while GPT-4 can handle up to 8,192 or even 32,768 tokens in some versions. This limit is an important constraint in LLM design and usage; if you exceed this limit, your requests will be rejected. Additionally, enterprise subscription models (e.g., Azure OpenAI) may offer larger context windows or vary by region, allowing for more extensive interactions in specific use cases.

The Gemini models support larger context windows. For example, Gemini 3 Pro and Gemini 3 Flash can handle up to 1,048,576 tokens in input, with an expanded output capacity of 65,536 tokens. This far exceeds the token limits of typical GPT models. Other Gemini variants, such as Gemini 2.5 Flash-Lite and Gemini 2.5 Pro, also support a full 1,048,576-token input limit.

From a billing perspective, most LLM services charge users based on the *total number of tokens processed*—including both input tokens and output tokens. Therefore, longer context windows and larger token limits can lead to higher costs if fully utilized.

But how can we measure how many tokens we use? Quite often, we need a way to count the input tokens to make sure the response doesn't stop midway. While the concept of tokens is universal across LLMs, the specific tokenization algorithms can vary between AI providers. This means that the exact token count for a given piece of text might differ slightly depending on the model and provider you're using.

We can measure our tokens with the help of some open source libraries that calculate the tokens of a prompt together with any system messages we provide as context.